Quantum Mechanics I — Exercise Sheet 6

Symmetry, conservation and degeneracy

Variant: Questions only

Conventions, as in Lecture 6: a symmetry of H is a unitary V with $VH = HV$; the three-level system has $H = \varepsilon(|e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|)$, the observables Z and X, the rotations $V_\alpha = e^{-i\alpha Z}$ and the mirror M; on the ring, $Te_x = e_{x+1}$.

Exercise 1 (conservation when the Hamiltonian changes in time). Let $H(t)$ be a Hamiltonian that depends continuously on time, with the evolution $U(t,0)$ of Lecture 2, and let $A = \sum_a a\,E_a$ be an observable.

1. Show that $\frac{\mathrm{d}}{\mathrm{d}t}\big(U(t,0)^\dagger E_a U(t,0)\big) = \frac{\mathrm{i}}{\hbar}\,U(t,0)^\dagger\,[H(t),E_a]\,U(t,0)$, and conclude that the statistics of A are constant for every state and time if and only if $[A,H(t)] = 0$ at every time.
2. For a qubit with $H(t) = \frac{\hbar\nu t}{2\tau}\sigma_x$, $\nu,\tau > 0$, check that $U(t,0) = e^{-i\phi(t)\sigma_x}$ with $\phi(t) = \nu t^2/(4\tau)$. Prepare $|\uparrow\rangle$, and find the probability of $+1$ along z at the time t. Note that $[\sigma_z, H(0)] = 0$: what does this show?

Exercise 2 (field and strain together). Add both a field along the axis and a strain to the three-level system: $H' = H + bZ + \eta X$ with $b,\eta > 0$.

1. Find the levels of H'.
2. Show that neither the mirror M nor the rotations V_α with $\alpha \notin \pi\mathbb{Z}$ are symmetries of H', but that $V_\pi = \mathrm{diag}(-1,1,-1)$ is. Which conserved observable does it give?

Exercise 3 (the symmetries of a qubit). Let $H = c\mathbb{1} + \frac{\hbar\nu}{2}\,\mathbf{n}\cdot\boldsymbol{\sigma}$ with $\nu > 0$ and $\mathbf{n}$ a unit vector, and $E_\pm = \frac{1}{2}(\mathbb{1} \pm \mathbf{n}\cdot\boldsymbol{\sigma})$.

1. Show that every unitary $V = e^{i\beta_1}E_+ + e^{i\beta_2}E_-$, with β_1, β_2 real, is a symmetry of H, and that every symmetry is of this form.
2. Conclude that any two symmetries of H commute. When does a qubit Hamiltonian have two symmetries that do not commute?

Exercise 4 (the ring of three and of four sites, by hand). For the ring Hamiltonian $H = c\mathbb{1} - \kappa(T + T^\dagger)$, $\kappa > 0$, find all the levels and an eigenvector basis for $N = 3$ and for $N = 4$, by hand, and check that a level is degenerate, as § 6.7 predicts.

Exercise 5 (second order, and a field across the axis). Use the notation of § 6.6 for a simple level ε of H_0 with unit eigenvector u: $h = \|H_1\| \le g/4$, λ the eigenvalue of H within $g/2$ of ε, $F = \mathbb{1} - E$, and, on the range of F, $S = (\lambda - FHF)^{-1}$ and $R = (\varepsilon - FH_0F)^{-1}$.

1. (a) Show that $|\lambda - \varepsilon| \le h$. (b) Show that $\|S\| \le 2/g$ and $\|R\| \le 1/g$. (c) Show that $S - R = S\big((\varepsilon-\lambda)\mathbb{1} + FH_1F\big)R$, so that $\|S - R\| \le 4h/g^2$. (d) Conclude that the second-order formula of Remark 6.11 is within $4h^3/g^2$ of λ.
2. Show that for the lowest level of H_0 the second-order correction is ≤ 0, and < 0 exactly when $FH_1u \neq 0$.

3. In the three-level system, let $H_0 = H$ and $H_1 = cX_\perp$ with $c > 0$, where $X_\perp$ is the matrix of § 6.4's box. For the simple level 0 of e_0 ($g = \varepsilon$), find the first- and second-order shifts. Then use the mirror to find all the levels of $H + cX_\perp$ exactly, and compare.
4. What happens to the level ε to first and to second order?

Exercise 6 ($\star$ constant means).

1. Show that an observable A is conserved under a constant H if and only if its mean $\langle \psi_t, A\psi_t \rangle$ is constant in time for every initial state ψ.
2. Under $H + \eta X$, compute the mean of Z in the states $U_t e_0$ and $U_t e_+$. What do they show?